Shenzhen Hi-Link Electronic Co., Ltd.

HLK-M59 specifications

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1. Product Brief

HLK-M59 is a cost-effective embedded serial port WiFi dual-frequency transparent transmission module developed and produced by HI-LINK Electronics.

The product is an embedded module based on a universal serial interface and in line with network standards, with a built-in TCP/IP protocol stack, which can realize the conversion between the user serial port and the wireless network (WIFI) interface, and can connect the serial port and the network TCP/UDP Transparently transmit data between.

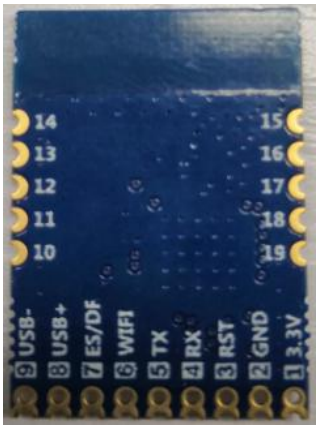
Through the HLK-M59 module, traditional serial devices can transmit their own data through the Internet without changing any configuration. A variety of networking methods and network connection types can be customized through simple configuration, providing a complete and fast solution for the user's serial device to transmit data through the network.

1.1. HLK-M59 module type

The module hardware type is shown below：



Pic 1 models front

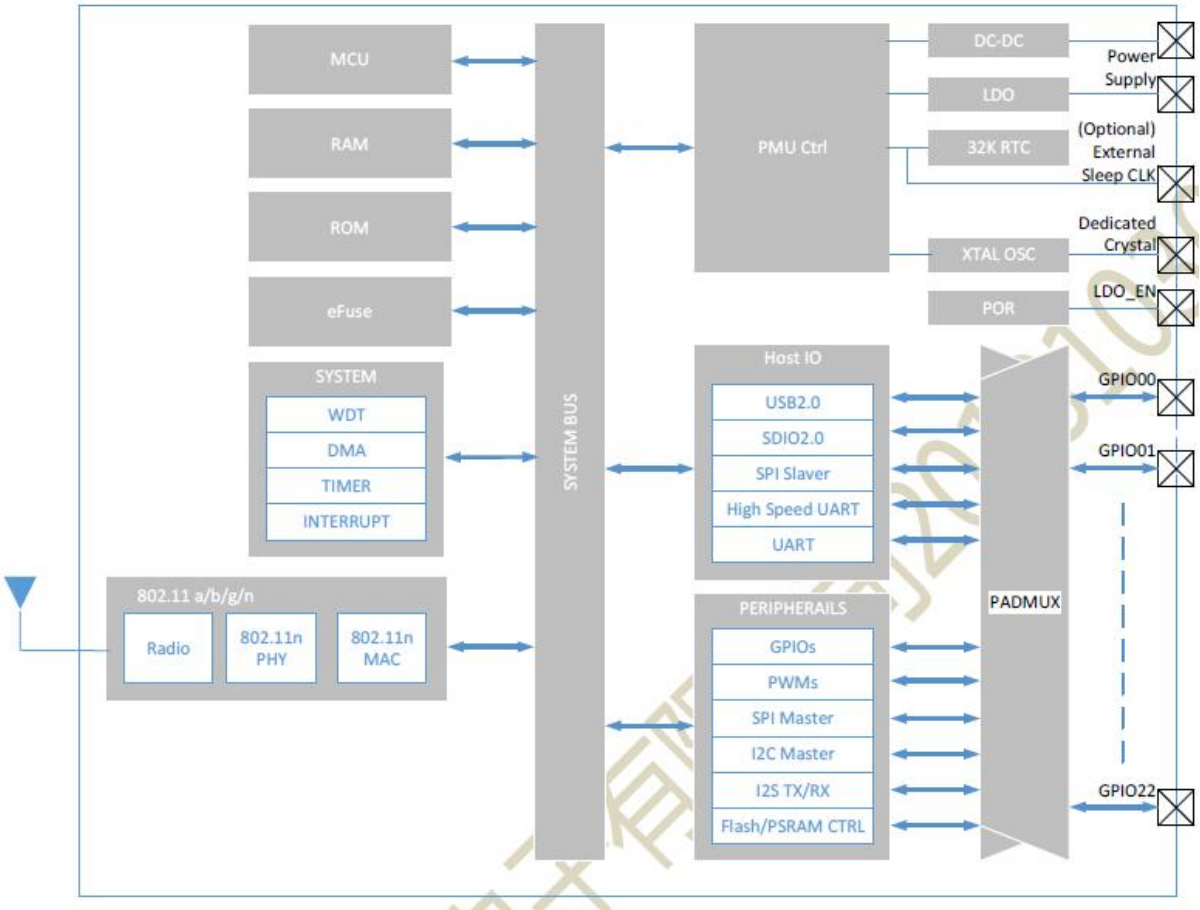


Pic 2 models on the back

1.2. MCU features

- Built -in 16Mbit flash
- Programmable function pin position
- GPIO that can be interrupted or awakened by external rising/falling edges
- Integrated UART×2/I2S×1/I2C×1/PWM×5/SPI×1/USB2.0×1
- Integrated 2 channel ADC
- Integrated watchdog and timer
- The operating system adopts freertos
- Support external psram interface

2. Functional block diagram



Pic 3 Module functional block diagram

3. Specifications

3.1. System memory

Project		Parameter
Memory system	Built-in RAM	192K Bytes User SRAM 8K Bytes icache
	Built-in Flash	Built-in 16Mbit flash

Form 1 System memory

3.2. Interface

Project	Parameter
GPIO	Support up to 8 GPIO port
Serial port performance	Support up to 2 serial ports, baud rate can be configured from 1200bps-300000bps
I2S interface performance	Support 2 channel I2S interface; BCLK of I2S host support 96/192/384/512/44.1/88.2KHz; support 16/32 bit per channel, data format can be configured as 16/20/24bit,Or determined by software (maximum 24bit per channel)
I2C interface performance	Support an I2C standard interface. Support master or slave operation
PWM interface performance	Support up to 5 PWM interfaces; PWM cycle and duty cycle are programmable, and the duty cycle can be configured between 0-100; cycle is programmable, software can select different clocks to generate long cycles
SPI interface	As SPI master, support up to 4 SPI slaves.The SPI clock is programmable and the maximum is 20Mhz; the data length can be configured through software, and the maximum is 64bit;
SDIO	Support 1 SDIO interface

Form 2 interface introduction

3.3. Power supply

3.3.1. Electrical specifications

Electrical parameters	
Power input voltage	DC:3.3±0.2V
I/O Power input range	2.7-3.6V、≤10mA
No-load running current	70±50mA
Average power consumption of the module	Around 250mW
Peak module current	370mA
Supply current requirements	≥500mA

Form 3 electrical parameters

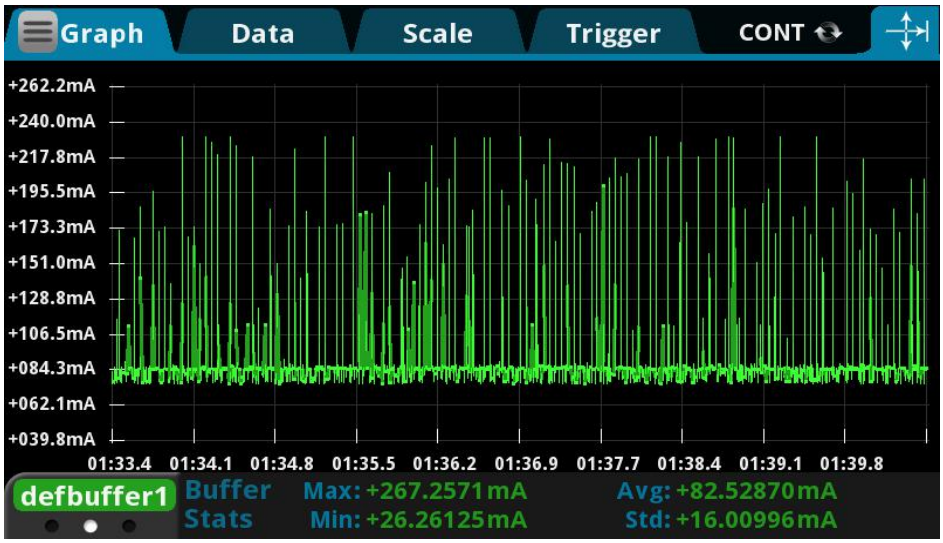
3.3.2. Current waveform

Module test environment: AP+2.4GSTA: module onboard antenna, equipment 3.3v power supply, single module test, the peak current of the module above the figure shall prevail.

5GSTA+ default mode: module onboard antenna, backplane power supply 5V, with backplane test, the peak current of the module above the figure shall prevail.

AP mode

3.3V power supply, the module is configured into AP mode to test the current, average value: 82mA, maximum value: 267mA. The detailed current waveform diagram is shown below.

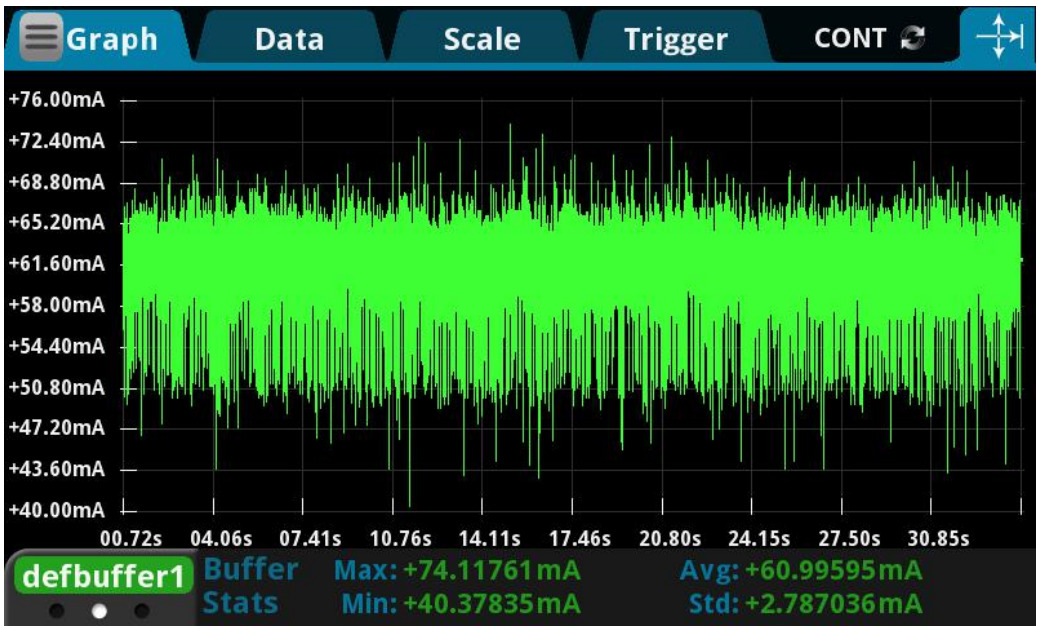


2.4G STA mode

3.3V power supply, the module is configured to 2.4gSTA mode test current, average value: 72mA, maximum value: 269mA. The detailed current waveform is shown below

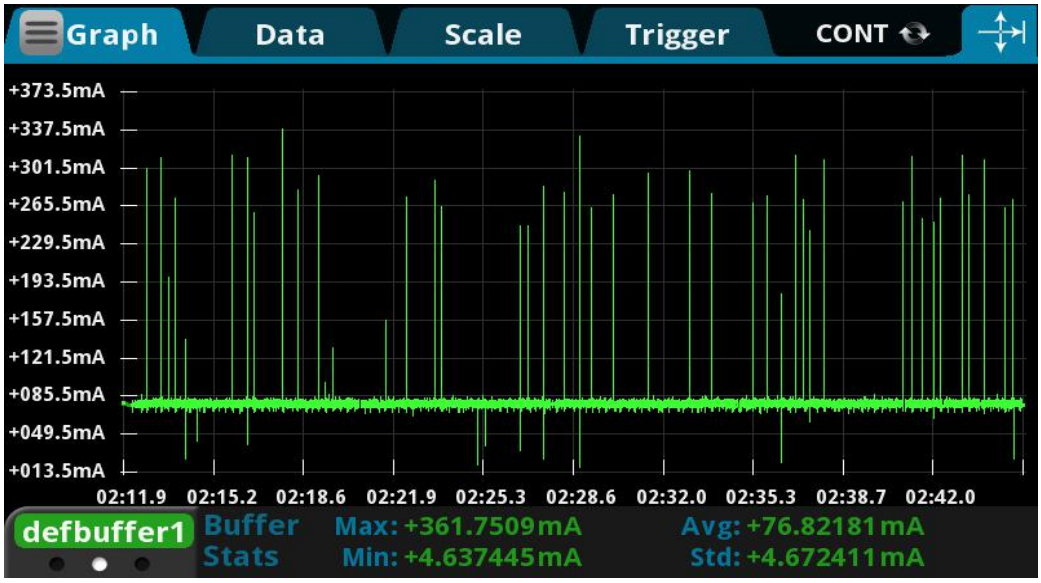
Default mode

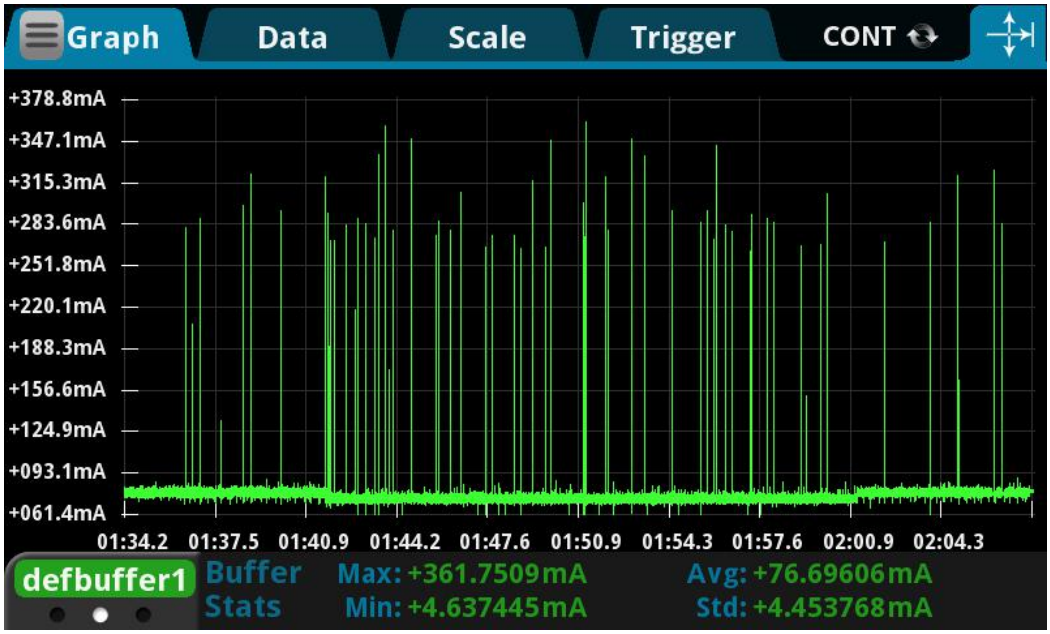
5V power supply with backplane, no module configuration, current obtained in mode test, average value: 60mA, maximum value: 74mA. The detailed current waveform is shown below。



5.8G STA

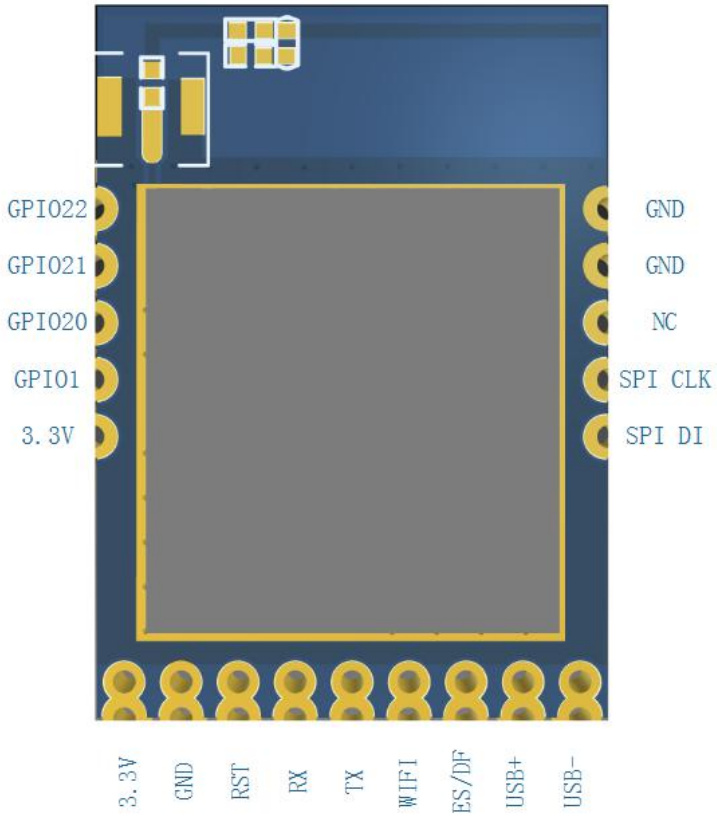
With 5V power supply on the bottom board, the module is configured into 5.8gAP mode to test the current, the average value: 76mA, the maximum value: 361mA. The detailed current waveform is shown below.





4. Module pin definition

4.1. Pin definition diagram



Pic 4 Pin definition diagram

Notes：

1，The picture is defined by default, the multiplexing function is not listed。

2，Please leave the unused functions. do not pull up and down at will, so as not to cause the module to start abnormally

4.2. Pin description

Pin number	Pin name	Type	Direction
1	**power**	POWER	power supply
2	**GND**	GND	buck ground
3	**RST**	-	reset
4	**RX**	I/O	UART0_RX
5	**TX**	I/O	UART0_TX
6	**WIFI**	O	WIFI LED STATUS
7	**ES/DF**	I	ES/DF KEY
8	**USB+**	I/O	USB positive signal
9	**USB-**	I/O	USB negative signal
10	**SPI DI**	I/O	USB MODE CHOOCE
11	**SPI CLK**	I/O	USB MODE CHOOCE
12	**NC**	NC	--
13	**GND**	GND	buck ground
14	**GND**	GND	buck ground
15	**GPIO22**	I/O	General purpose input/output
16	**GPIO21**	I/O	General purpose input/output
17	**GPIO20**	I/O	General purpose input/output
18	**GPIO1**	I/O	General purpose input/output
19	**power**	POWER	General purpose input/output

Form 4　Pin description

Pin type definition：

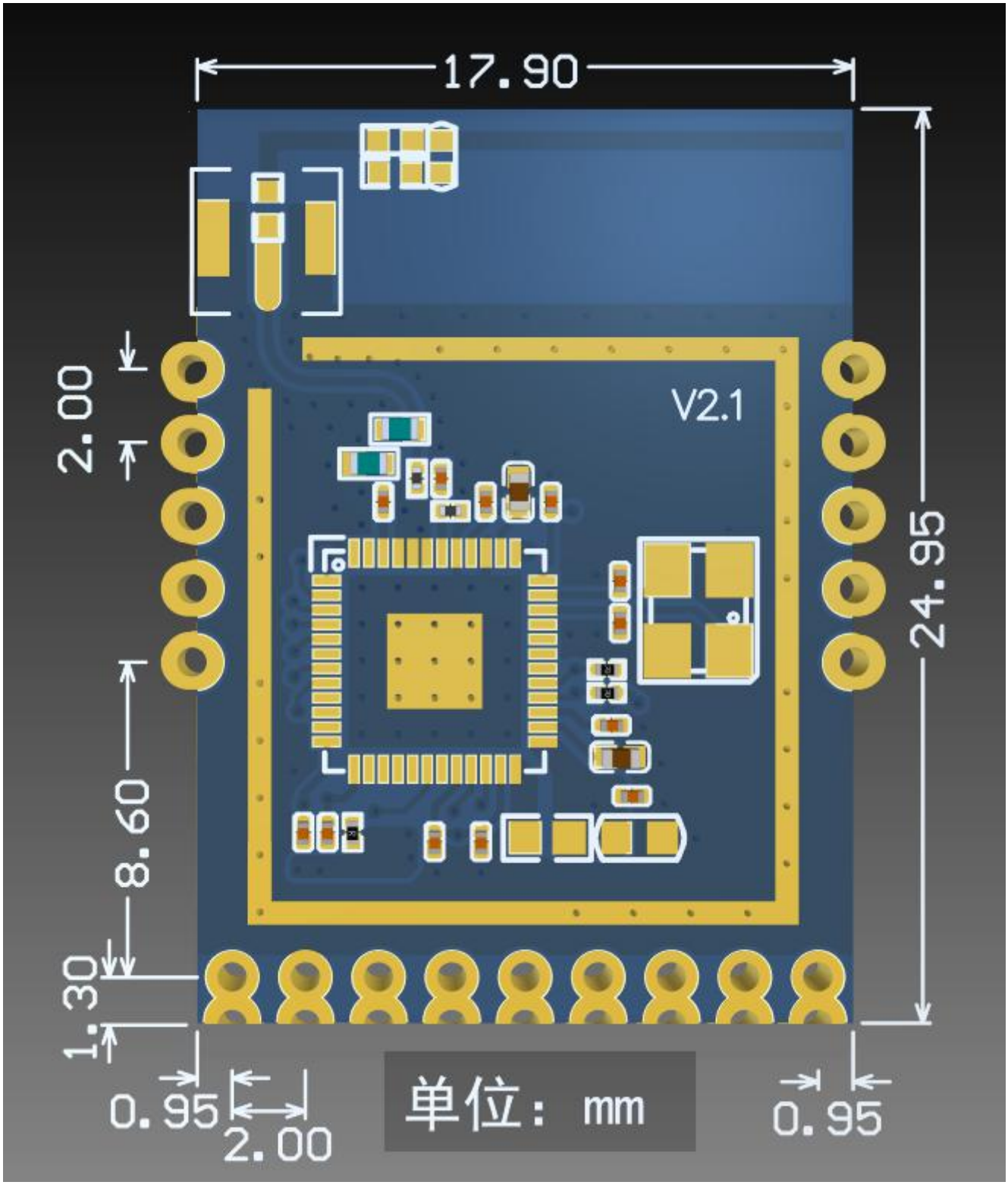
- *I/O → Digital input/output;*
- *I → Digital input*
- *O → Digital output*
- *A, I → Analog input*
- *A, O → Analog output*
- *A, I/O → Analog input/output*
- *PWR → Power*
- *GND → Ground*

4.3.GPIO alternate function and function pin location

	ALT0	ALT1	ALT2	ALT3	ALT4	ALT5
P00		PWM0			ADC3	GPIO0
P01		PWM1				GPIO1
P02		PWM2				GPIO2
P03	UART0	PWM3				GPIO3
P04	UART0	PWM4				GPIO4
P05	UART1	I2S				GPIO5
P06	UART1	I2S				GPIO6
P07	UART1	I2S				GPIO7
P08		SPIM	I2S			GPIO8
P09	UART1	I2S				GPIO9
P10		SPIM	I2S	I2CM	SPIS	GPIO10
P11					SPIS	GPIO11
P12		SPIM	I2S	I2CM	SPIS	GPIO12
P13		SPIM	I2S	PSRAM	SPIS	GPIO13
P14	FLASH			PSRAM		GPIO14
P15	FLASH			PSRAM		GPIO15
P16	FLASH			PSRAM		GPIO16
P17	FLASH			PSRAM		GPIO17
P18	FLASH			PSRAM		GPIO18
P19	FLASH			PSRAM		GPIO19
P20		I2S	I2S		ADC2	GPIO20
P21	I2CS	UART0	I2CM		ADC1	GPIO21
P22	I2CS	UART0	I2CM		ADC0	GPIO22

Form 5 Pin multiplexing

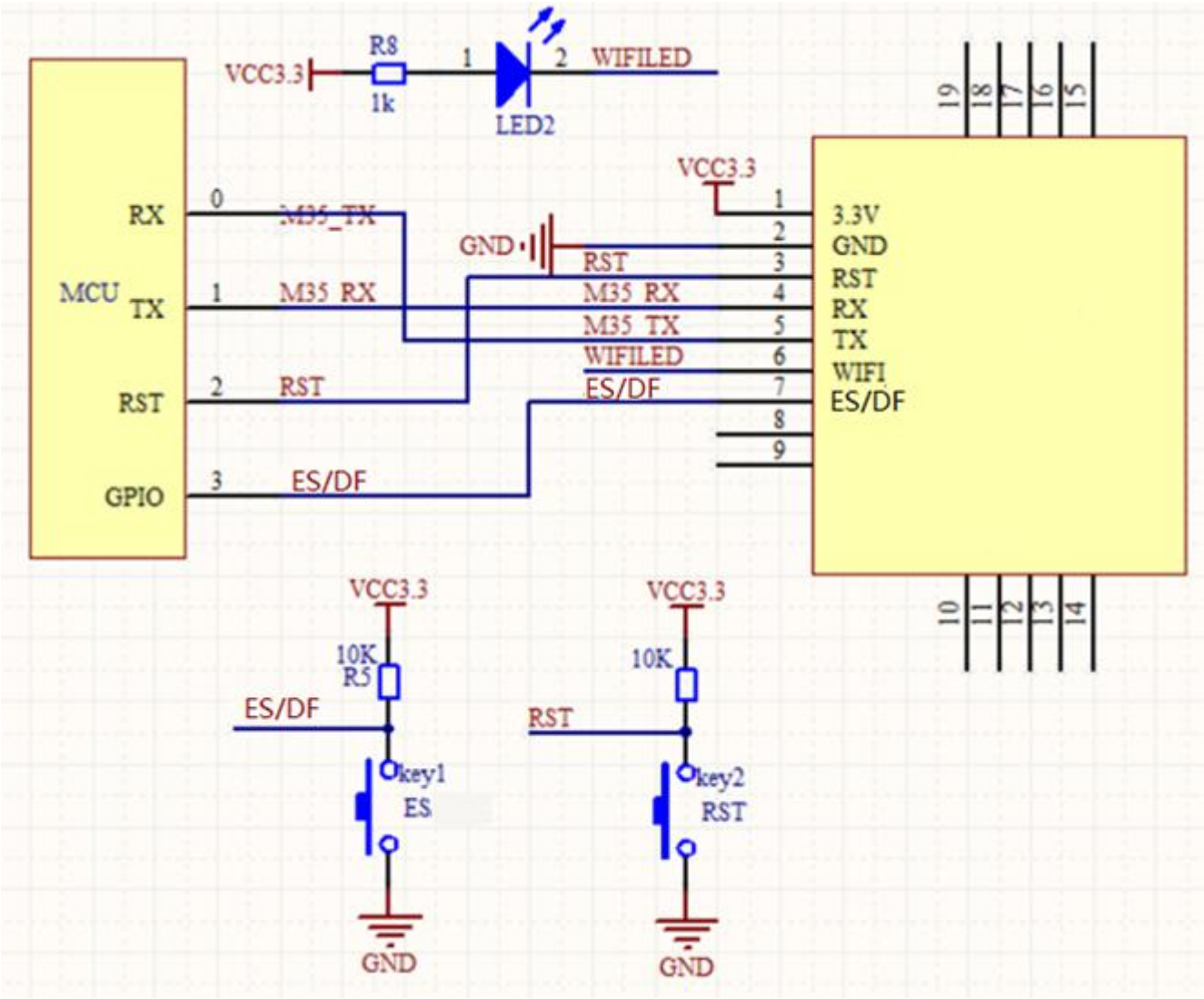
5. Module dimension drawing



unit：mm

Pic 5　Dimensional drawing of the module

6. Typical application circuit

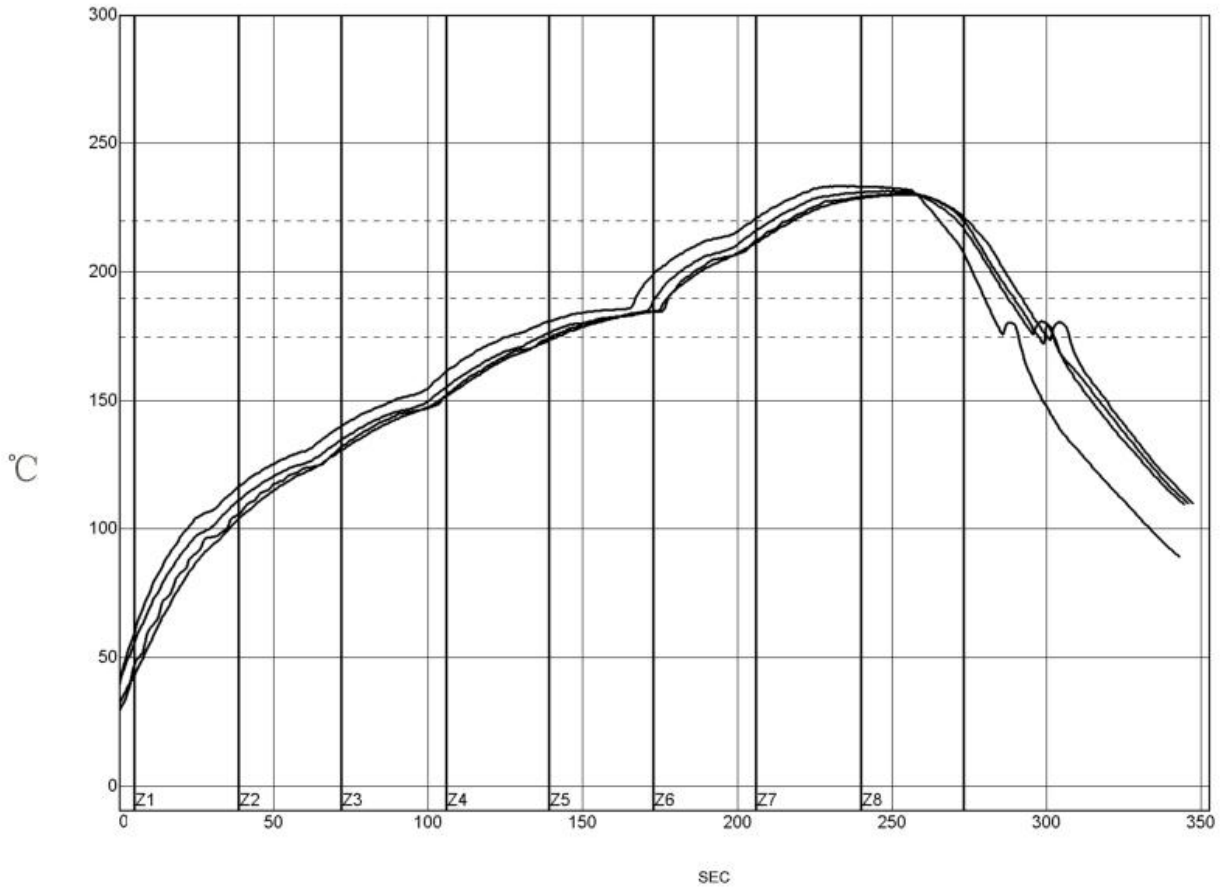


Pic 6 Test board schematic

7. Recommended reflow temperature

When the module passes the furnace for the second time, please strictly follow this temperature curve. If the temperature deviation of reflow soldering is too higher, the module will be damaged!.

Temperature zone	1	2	3	4	5	6	7	8
Upper temperature zone	125	135	155	185	195	225	240	230
Lower temperature zone	125	135	155	185	195	225	240	230



PWI= 94%	Constant temperature time 175 to 190℃		Reflow time/220℃		Maximum temperature	
<TC2>	35.53	-82%	55.58	-72%	230.28	-94%
<TC3>	37.66	-74%	58.66	-57%	230.56	-89%
<TC4>	41.52	-62%	60.63	-47%	233.62	-28%
<TC5>	37.07	-76%	60.44	-48%	231.67	-67%
温差	5.99		5.05		3.34	

Process interface

Solder paste:	System Default for Reflow		
Statistics name	Minimum interface	Highest interface	unit
Constant temperature time 175-190℃	30	90	sec
Time above reflow -220℃	50	90	sec
Maximum temperature	230	240	℃

8. Revisition

Date	Version	Revised content
2020/3/4	1.0	initial version
2020/3/19	1.1	Increase electrical parameters
2020/3/25	1.2	Update picture
2020/4/10	1.3	Update current waveform
2020/4/25	1.4	Update module 12 pin description